

Experiment 33: Number of Vowels

Aim

To write and execute a **Prolog program to find the number of vowels in a given word or list of characters.**

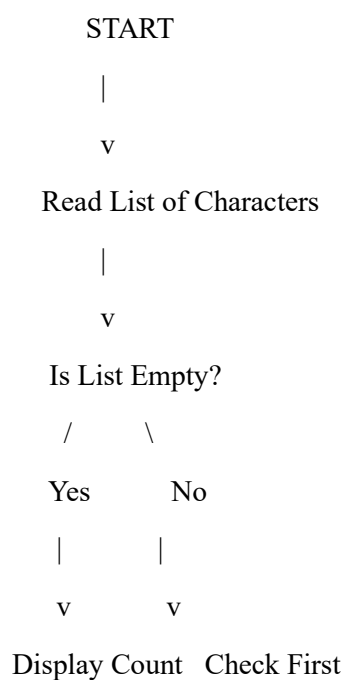
Objectives

1. To identify vowels using Prolog predicates.
2. To count vowels present in a given list.
3. To understand recursion in Prolog.
4. To practice list processing.
5. To obtain the total number of vowels in a word.

Algorithm

1. Start the program.
2. Define the vowels as a, e, i, o, u.
3. Accept a list of characters as input.
4. Check the first character.
5. If it is a vowel, increment the count.
6. If it is not a vowel, keep the count unchanged.
7. Recursively process the remaining characters.
8. When the list becomes empty, return the final count.
9. Stop.

Flowchart



```

Character
|
v
Is it a Vowel?
/      \
Yes      No
|        |
v        v
Count = Count+1  Count unchanged
\              /
\              /
v              v
Process Rest
|
v
List Empty?
|
v
Display Count
|
v
END

```

Program

```

% Vowels
vowel(a).
vowel(e).
vowel(i).
vowel(o).
vowel(u).

```

```

% Base case
count_vowels([], 0).

```

% If the first character is a vowel

count_vowels([H|T], Count) :-

 vowel(H),

 count_vowels(T, C),

 Count is C + 1.

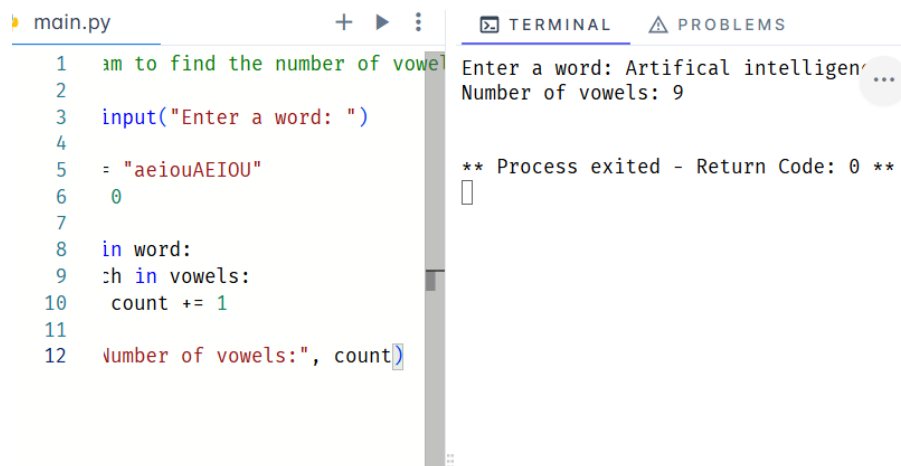
% If the first character is not a vowel

count_vowels([_|T], Count) :-

 \+ vowel(H),

 count_vowels(T, Count).

Output



The screenshot shows a code editor with a file named 'main.py'. The code is a Python script designed to count vowels in a word. It includes a comment 'I am to find the number of vowels in a given list of characters', a function definition 'def count_vowels(word):', a list of vowels 'vowels = "aeiouAEIOU"', and a loop that iterates through each character in the word, incrementing a counter if the character is in the vowels list. The output window on the right shows the program being run, with the input 'Artificial intelligence' and the resulting output 'Number of vowels: 9'. The process exited with a return code of 0.

```
main.py
1  I am to find the number of vowels in a given list of characters
2
3  def count_vowels(word):
4
5      vowels = "aeiouAEIOU"
6      count = 0
7
8      for char in word:
9          if char in vowels:
10             count += 1
11
12     return count
```

TERMINAL

Enter a word: Artificial intelligence
Number of vowels: 9

** Process exited - Return Code: 0 **

Result

The Prolog program was successfully executed to find the number of vowels in a given list of characters.

Conclusion

Thus, the number of vowels was successfully calculated using **Prolog predicates, recursion, and list processing**. The program correctly identifies vowels and produces their total count.